Understanding Solution of Triangle - Made Simple

Introduction: What is a Triangle and What Does "Solution" Mean?

A **triangle** is a shape with three sides and three angles. In this chapter, we learn how to find the missing sides and angles when we already know some of them. To **"solve"** a triangle simply means to find all the missing information about that triangle.

Think of it like a puzzle. If someone gives you three pieces of information about a triangle (like two sides and one angle), you can figure out the rest of the triangle. That's what we mean by "solving" a triangle.

A triangle has **six elements in total**:

- Three sides (let's call them a, b, and c)
- Three angles (let's call them A, B, and C)

When we know three of these elements, we can usually find the other three. That's what this chapter teaches us.

The Two Main Rules: Sine Rule and Cosine Rule

To solve triangles, we use two special formulas that show the relationship between sides and angles. Let me explain both of them in the simplest way possible.

What is the Sine Rule?

The **Sine Rule** says that if you divide each side by the sine of its opposite angle, you always get the same number. Think of "opposite angle" as the angle across from that side.

In symbols, this looks like:
$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

What does this mean? It means these three ratios are always equal. This rule is super helpful when you know some angles and sides, and you want to find other sides and angles.

When to use the Sine Rule:

- When you know two angles and one side
- When you know two sides and one angle (but NOT the included angle - meaning NOT the angle between those two sides)

What is the Cosine Rule?

The **Cosine Rule** (also called Cosine Formula) is a bit more complex, but it's like the Pythagorean theorem's bigger cousin.

The Cosine Rule says:
$a^2 = b^2 + c^2 - 2bc\cos A$

Or you can rearrange it to find an angle:
$\cos A = \frac{b^2+c^2-a^2}{2bc}$

What does this mean? It shows the relationship between all three sides and one angle. This rule is perfect when you have information about all three sides, or when you have two sides and the angle between them.

When to use the Cosine Rule:

- When you know all three sides of the triangle
- When you know two sides AND the angle between them (we call this the "included angle")

Five Different Cases of Triangle Solutions

Now let's look at the five different situations you might face when solving a triangle. Each situation needs a slightly different approach.

Case 1: When You Know All Three Angles

What you're given: All three angles (A, B, and C)

Important note: This is a special case. You CANNOT find a unique triangle because there are infinite triangles with the same three angles but different sizes. Imagine drawing a triangle with angles of 60°, 60°, and 60° on a piece of paper. Now imagine drawing the same triangle twice as big. Both triangles have the same angles!

What you can do: Using the Sine Rule, you can find the RATIO of the sides:
$a : b : c = \sin A : \sin B : \sin C$

This tells you the shape of the triangle, but not how big it is. Unless someone tells you the actual length of one side, you can't find the exact size.

Example: If all angles are given, you know the triangle's shape but not its size.

Case 2: When You Know All Three Sides

What you're given: All three sides (a, b, and c)

How to solve it: Use the Cosine Rule to find each angle:
$\cos A = \frac{b^2+c^2-a^2}{2bc}$

$\cos B = \frac{a^2+c^2-b^2}{2ac}$
$\cos C = \frac{a^2+b^2-c^2}{2ab}$

Steps:

1. Use the Cosine Rule formula above to find angle A
2. Use it again to find angle B
3. Find angle C using the fact that all angles must add up to 180°: C = 180° - A - B

Check your work: Make sure A + B + C = 180°

Case 3: When You Know Two Angles and One Side

What you're given: Two angles (say A and B) and one side (say a)

How to solve it:

1. First, find the third angle: C = 180° - A - B (remember, all angles must add up to 180°)
2. Now use the Sine Rule to find the other two sides:
   $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Steps:

- Find b using: $b = \frac{a \sin B}{\sin A}$
- Find c using: $c = \frac{a \sin C}{\sin A}$

This is the easiest case because there's always exactly one solution - no confusion!

Case 4: When You Know Two Sides and the Included Angle

What you're given: Two sides (say b and c) and the angle between them (say A)

Important: The angle MUST be between the two sides you know. This is called the "included angle."

How to solve it:

Step 1: Find the third side using the Cosine Rule:
$a^2 = b^2 + c^2 - 2bc \cos A$

Take the square root to find a.

Step 2: Find another angle using the Sine Rule:
$\sin B = \frac{b \sin A}{a}$

Step 3: Find the last angle:
$C = 180^\circ - A - B$

This case also has exactly one solution - straightforward and simple!

Case 5: When You Know Two Sides and an Opposite Angle (The Tricky Case!)

What you're given: Two sides (say b and c) and an angle opposite to ONE of them (say B, which is opposite side b)

Important note: The angle is NOT between the two sides. This is called the "non-included angle" or "opposite angle." This is the trickiest case!

Why is it tricky? This situation can have:

- NO solution (no triangle exists)
- ONE solution (exactly one triangle exists)
- TWO solutions (two different triangles exist!)

This is why mathematicians call it the **"Ambiguous Case"** (ambiguous means unclear or having multiple meanings).

How to figure out what happens:

Step 1: Use the Sine Rule to find angle C:
$\sin C = \frac{c \sin B}{b}$

Step 2: Now check what happens:

- **If sin C > 1:** This is impossible (sine can never be bigger than 1), so NO TRIANGLE EXISTS
- **If sin C = 1:** Exactly ONE triangle exists, and angle C = 90°
- **If sin C < 1:** You get two possible values for C (one acute angle and one obtuse angle)
  - For the acute angle: $C_1 = \sin^{-1}(\sin C)$
  - For the obtuse angle: $C_2 = 180° - \sin^{-1}(\sin C)$
  - However, you need to check if both actually work

Important sub-rules for when angle B is acute:

- If b < c and b < height (where height = c sin B), then NO TRIANGLE exists
- If b ≥ c, then ONE triangle exists
- If c > b > height, then TWO triangles exist

Important sub-rules for when angle B is obtuse:

- If b ≤ c, then NO TRIANGLE exists
- If b > c, then ONE triangle exists

Why does this happen? Imagine you have two sticks of different lengths and you're trying to connect them with a third stick at a specific angle. Depending on their lengths, you might not be able to form a triangle, or you might be able to form it in two different ways!

Step-by-Step Method to Identify Which Case You Have

When you get a problem, follow these steps:

Step 1: Count what you know

- Count the sides you know
- Count the angles you know

Step 2: Identify the case:

- Do you know all 3 angles? → **Case 1**
- Do you know all 3 sides? → **Case 2**
- Do you know 2 angles and 1 side? → **Case 3**
- Do you know 2 sides and the angle between them? → **Case 4**
- Do you know 2 sides and an angle NOT between them? → **Case 5** (the ambiguous case)

Step 3: Apply the correct method from above

Quick Reference: Which Rule to Use

Situation	What You Know	What to Use	Number of Solutions
Case 1	3 angles	Sine Rule for ratios	Shape only, no size
Case 2	3 sides	Cosine Rule	Always 1
Case 3	2 angles + 1 side	Sine Rule	Always 1
Case 4	2 sides + included angle	Cosine Rule then Sine Rule	Always 1
Case 5	2 sides + opposite angle	Sine Rule	0, 1, or 2

Important Formulas Summary

Sine Rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine Rule (for sides):

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Cosine Rule (for angles):

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

Angle Sum:

$A + B + C = 180^\circ$

Real-World Example

Let's say you're a builder and you have a triangular piece of land. You measure:

- Side a = 50 meters
- Side b = 60 meters
- Side c = 70 meters

You want to find all the angles.

Which case is this? Case 2 (all three sides known)

Solution:
Use the Cosine Rule:
$\cos A = \frac{60^2+70^2-50^2}{2(60)(70)} = \frac{3600+4900-2500}{8400} = \frac{6000}{8400} = 0.714$

So A = $\cos^{-1}(0.714) \approx 44.4°$

You would repeat this for angles B and C.

Key Things to Remember

1. **There are six elements:** three sides and three angles
2. **To solve a triangle:** You need at least three of these elements (with at least one being a side)
3. **Two main rules:** Sine Rule and Cosine Rule
4. **Five different situations:** Each has its own approach
5. **The tricky case:** When you have two sides and an opposite angle (it can have 0, 1, or 2 solutions)
6. **Always check:** Make sure your angles add up to 180°

Conclusion

Solving triangles might seem complicated at first, but once you identify which case you're dealing with, the method becomes clear. The key is to:

1. Know what information you have
2. Identify which case it is
3. Apply the correct formula
4. Solve step by step

With practice, this will become second nature. Remember, in Case 5 (the ambiguous case), always be careful to check if one solution, two solutions, or no solution exists. For all other cases, you should get

exactly one answer.

Good luck with your learning!

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